Masters

Parallel and Distributed Computing (CS621)

Question: 1 (Marks: 1) Attempted Questions: 0 Total Questions: 50

In _____ control strategy, the control function is distributed among different components in the system.

Choices:

- decentralized

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- distributed
- centralized
- supervised

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Parallel and Distributed Computing (CS621)

Question: 2 (Marks: 1) Attempted Questions: 1 Total Questions: 50

In _____ switching, a complete path has to be established prior to the start of communication between a source and a destination.

Choices:

- packet
- port
- circuit
- connect

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Parallel and Distributed Computing (CS621)

Question: 3 (Marks: 1) Attempted Questions: 2 Total Questions: 50

Number of bits transmitted in a unit of time is known as_____.

Choices:

latency

diameter

bandwidth



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bisection

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Parallel and Distributed Computing (CS621)

Question: 4 (Marks: 1) Attempted Questions: 3 Total Questions: 50

A _____ topology is a network interconnection scheme in which each node is connected to every other node with a direct link.

Choices:

- bus
- star
- mesh
- switch

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Parallel and Distributed Computing (CS621)

Question: 5 (Marks: 1) Attempted Questions: 4 Total Questions: 50

The possibility to use the same software in different environments is known as _____.

Choices:

reliability

portability

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robustness

availability

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Parallel and Distributed Computing (CS621)

Question: 10 (Marks: 1) Attempted Questions: 6 Total Questions: 50

Decomposition of computation into a large number of tasks results in _____ decomposition.

Choices:

- fine-grained

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- course-grained
- medium-grained
- best-grained

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Parallel and Distributed Computing (CS621)

Question: 11 (Marks: 1) Attempted Questions: 7 Total Questions: 50

Discrete optimization problems can be solved by using _____ decomposition technique.

Choices:

- Recursive decomposition
- Data decomposition
- Exploratory decomposition
- Speculative decomposition

Correct

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Parallel and Distributed Computing (CS621)

Question: 12 (Marks: 1) Attempted Questions: 8 Total Questions: 50

The problem of finding the minimum in a list is an example of decomposition of data based on _____

Choices:

- output data
- input data
- intermediate data
- both input and output data

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Parallel and Distributed Computing (CS621)

Question: 13 (Marks: 1) Attempted Questions: 9 Total Questions: 50

Which of the following decomposition techniques is/are more desirable for quicksort problem?

Choices:

- Recursive decomposition
- Data decomposition
- Exploratory decomposition
- Both data and recursive decomposition

correct

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Parallel and Distributed Computing (CS621)

Question: 26 (Marks: 1) Attempted Questions: 10 Total Questions: 50

Sequential Consistency ensures that the result of any execution in a distributed system is equivalent to:

Choices:

- Operations being executed in a random order
- All processes executing concurrently without synchronization
- Read and write operations being executed in a sequential order
- Each process executing independently of others

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Parallel and Distributed Computing (CS621)

Question: 27 (Marks: 1) Attempted Questions: 11 Total Questions: 50

What does causal consistency guarantee regarding writes in a distributed system?

Choices:

- All writes are seen by all processes in the same order.
- Concurrent writes are seen in the same order on different machines.
- All writes are executed concurrently to improve performance.
- Writes that are causally related are seen in the same order by all processes.

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Parallel and Distributed Computing (CS621)

Question: 29 (Marks: 1) **Attempted Questions: 13** **Total Questions: 50**

GPUs have gained popularity due to their ability to:

Choices:

- Run only real-world applications efficiently.
- Execute single-threaded tasks faster than multi-core CPUs.
- Replace multi-core CPUs in all computing tasks.
- Provide parallel processing capabilities for certain applications.

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Parallel and Distributed Computing (CS621)

Question: 30 (Marks: 1) **Attempted Questions: 14** **Total Questions: 50**

CUDA is designed for computation-intensive tasks to be executed on which hardware?

Choices:

- CPU (Central Processing Unit)
- GPU (Graphics Processing Unit)
- RAM (Random Access Memory)
- SSD (Solid State Drive)

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Parallel and Distributed Computing (CS621)

Question: 31 (Marks: 1) Attempted Questions: 15 Total Questions: 50

GPUs accelerated libraries are commonly used in which type of applications?

Choices:

- Text editors
- Web browsers
- Data science and machine learning
- File compression utilities

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Parallel and Distributed Computing (CS621)

Question: 32 (Marks: 1) Attempted Qu

In message passing, how do processors exchange data in a distributed memory system?

Choices:

- They share a global memory space.
- They use cache synchronization mechanisms.
- They communicate through a communication network.
- They directly access each other's local memory.

Masters

Question: 34 **(Marks: 1)** **Attempted Questions:** 18 **Total Questions:** 50

What is the size of the Basic MPI library in terms of the number of functions?

Choices:

- ○ Large (over 50 functions)
- ○ Medium (around 25 functions)
- ○ Small (around 10 functions) Masters
- ○ Basic (6 functions) Correct

Parallel and Distributed Computing (CS621)

Question: 35 (Marks: 1) Attempted Questions: 19 Total Questions: 50

In MPI, which function is commonly used to send a message from one process to another?

Choices:

- MPI_Broadcast
- MPI_Reduce
- MPI_Send
- MPI_Receive

Masters

Parallel and Distributed Computing (CS621)

Question: 36 (Marks: 1) Attempted Questions: 20 Total Questions: 50

MapReduce is a software framework which supports _____computing on large data sets.

Choices:

- serial
- parallel
- distributed
- both parallel and distributed

Masters

Parallel and Distributed Computing (CS621)

Question: 37 (Marks: 1) Attempted Questions: 21 Total Questions: 50

Which of the following is NOT true regarding Google File System (GFS)?

Choices:

- High security
- Limited security
- Only viable in a specific environment
- Custom designed

Masters

Parallel and Distributed Computing (CS621)

Question: 38 (Marks: 1) Attempted Questions: 22 Total Questions: 50

_____ of Hadoop Distributed File System (HDFS) focuses on the individual instances of application data or the data content.

Choices:

Structural metadata Masters

Descriptive metadata Correct

Contextual metadata

Contextual and structural metadata

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 39 (Marks: 1) **Attempted Questions: 23** **Total Questions: 50**

What is the primary function of Hadoop's HDFS (Hadoop Distributed File System)?

Choices:

- Running complex machine learning algorithms.
- Storing and distributing data across a cluster of machines.
- Managing user access control.
- Creating graphical visualizations

Masters

Parallel and Distributed Computing (CS621)

Question: 40 (Marks: 1) Attempted Questions: 24 Total Questions: 50

Which component of Hadoop is responsible for storing and managing data across a cluster?

Choices:

- Hadoop Mapper
- Hadoop Reducer
- Hadoop NameNode
- Hadoop TaskTracker

Masters

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 41 (Marks: 3) Attempted Questions: 25 Total Questions: 50

Define the degree of the switch in interconnection networks.

.....

Answer:

The total number of ports on a switch is the degree of the switch. Switches may also provide support for internal buffering, routing and multicast.

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Parallel and Distributed Computing (CS621)

Question: 42 (Marks: 3) Attempted Questions: 26 Total Questions: 50

Bill Dally, a prominent computer scientist has highlighted power efficiency and massive parallelism as major advantages of GPUs by considering two aspects. You are required to mention these aspects.

Answer:

Bill Dally of Stanford University considers power and massive parallelism as the major benefits of GPUs over CPUs for the future.

Two Aspects:

1. **Data Access Rate Capability**
   a. **Bandwidth**
2. **Data Processing Capability**
   a. **How many ops per sec**

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Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 43 (Marks: 3) Attempted Questions: 28 Total Questions:

Write down the names of the data-centric consistency models.

Answer:

Data-centric Consistency Models:

I. Continuous Consistency
II. Consistent Ordering of Operations
III. Causal Consistency
IV. Grouping Operations

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Parallel and Distributed Computing (CS621)

Question: 44 (Marks: 3) Attempted Questions: 28 Total Questions: 50

What will happen when multiple threads attempt to manipulate the same data item, if proper care is not taken to synchronize?

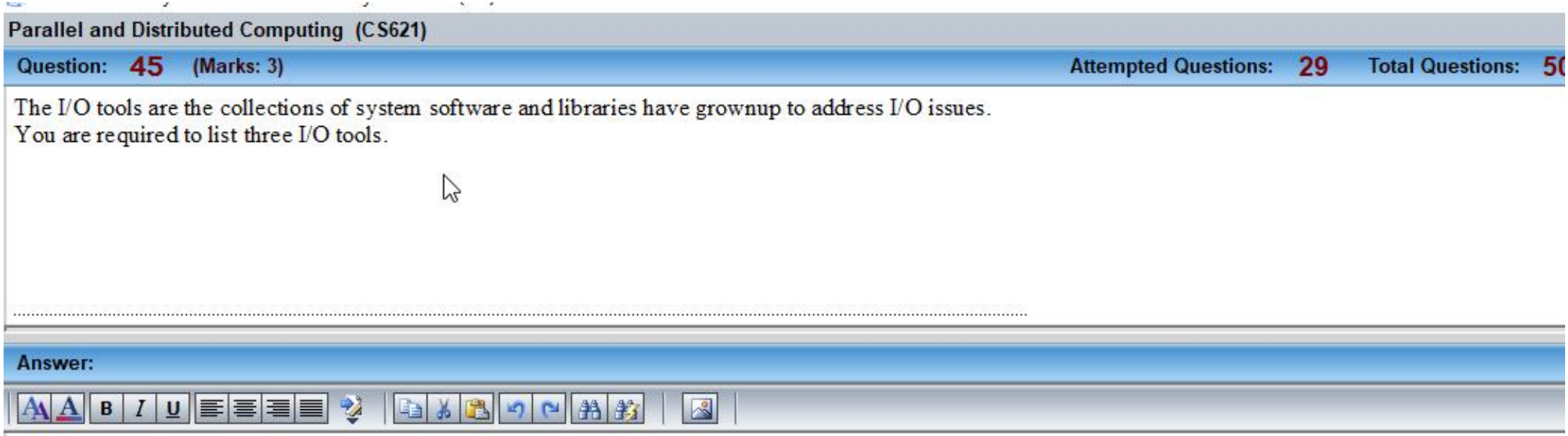
Can results incoherent

Answer:

When multiple threads attempt to manipulate the same data item without proper synchronization, results can become incoherent or inconsistent

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1. Parallel File Systems (e.g., Lustre, GPFS, and HDFS)
2. MPI-IO (Message Passing Interface for Input/Output)
3. High level libraries (e.g., HDF5 and NetCDF)

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Parallel and Distributed Computing (CS621)

Question: 46 (Marks: 3) Attempted Questions: 30 Total Questions:

Do you think, a parallel runtime of a program can be evaluated asymptotically as a function of input size only? Justify your answer in either case.

Answer:

The parallel runtime of a program depends on the input size, the number of processors, and the communication parameters of the machine.

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

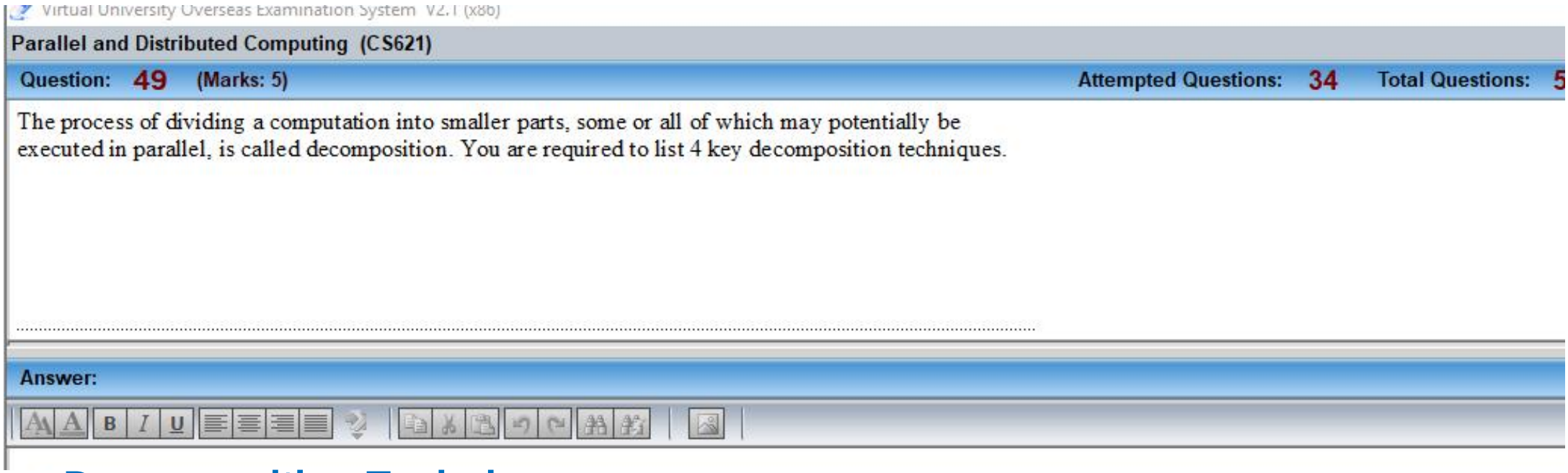
Question: 47 (Marks: 5) **Attempted Questions: 31**

Write down any five interconnection network's properties in parallel and distributed computing.

.....

Answer:

1.Topology: The arrangement of nodes and links in the network.
2. Diameter: The maximum distance between any two nodes in the network.
3. Degree: The number of direct connections a node has.
4. Bandwidth: The data transfer capacity of the network links.
5. Fault Tolerance: The ability of the network to function correctly even when some components fail.



Decomposition Techniques:

i. Recursive Decomposition
ii. Data Decomposition
iii. Exploratory Decomposition
iv. Speculative Decomposition

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Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 50 (Marks: 5) **Attempted Questions: 34** **Total Questions: 50**

Differentiate the following statement as Independent MPI IO and Collective MPI IO calls.

1. MPI_File_write()
2. MPI_File_write_all()
3. MPI_File_read().
4. MPI_File_read_all()

Answer:

1. MPI_File_write()
Type: Independent MPI I/O

2. MPI_File_write_all():
Type: Collective MPI I/O

3. MPI_File_read():
Type: Independent MPI I/O

4. MPI_File_read_all():
ype: Collective MPI I/O

Parallel and Distributed Computing (CS621)

Question: 7 (Marks: 1) Attempted Questions: 35 Total Questions: 50

Much of the effort associated with writing correct threaded programs is spent on _____ concurrent threads.

Choices:

- scheduling
- data accesses for
- synchronizing
- both scheduling and data access

Masters

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 8 **(Marks: 1)** **Attempted Questions: 36** **Total Questions: 50**

Which of the following can be used to overcome the overhead of polling for availability of locks?

Choices:

- Semaphores
- Mutex-Locks
- Condition variable
- trylock () function

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Parallel and Distributed Computing (CS621)

Question: 9 (Marks: 1) Attempted Questions: 37 Total Questions: 50

The length of the _____in a task dependency graph is called the critical path length.

Choices:

- shortest path
- longest path
- average of longest and shortest path
- intersection of longest and shortest path

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Parallel and Distributed Computing (CS621)

Question: 13 (Marks: 1) Attempted Questions: 38

Which of the following decomposition techniques is/are more desirable for quicksort problem?

Choices:

- (●) Recursive decomposition
- ( ) Data decomposition
- ( ) Exploratory decomposition
- Both data and recursive decomposition

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Parallel and Distributed Computing (CS621)

Question: 14 (Marks: 1) **Attempted Questions: 38** **Total Questions: 50**

Keep in view of Lustre file system, Identify, which of the following provides file service, and network request handling for one or more local OSTs?

.....

Choices:

- Matadata Server (MDS)

.....

- Matadata Target (MDT)

.....

- Object Storage Server (OSS)

.....

- Object Storage Target (OST)

Masters

Parallel and Distributed Computing (CS621)

Question: 15 (Marks: 1) Attempted Questions: 39 Total Questions: 5(

Keep in view of Lustre file system, Identify, which of the following stores file data (chunks of files)?

Choices:

- Matadata Server (MDS)
- Matadata Target (MDT)
- Object Storage Server (OSS)
- Object Storage Target (OST)

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Parallel and Distributed Computing (CS621)

Question: 16 (Marks: 1) Attempted Questions: 40 Total Questions: 5

Which of the following is not true regarding POSIX I/O interface?

Choices:

- Standard I/O interface across many platforms
- open, read/write, close functions in C/C++/Fortran
- Mechanism almost all serial applications use to perform I/O
- Many ways for describing collective access

Masters

Virtual University Overseas Examination System V2.1 (x86)

Parallel and Distributed Computing (CS621)

Question: 17 (Marks: 1)

Which of the following is not true regarding MPI-IO?

Choices:

- Provides a high-level interface
- Has a large number of routines
- Facilitate Collective I/O
- Facilitate Atomicity rules

Masters

Parallel and Distributed Computing (CS621)

Question: 18 (Marks: 1) **Attempted Questions: 42** **Total Questions: 50**

Aggregates smaller read/write operations into larger operations is an example of_____.

.....

Choices:

- POSIX IO

  .....

- MPI IO

  .....

- Buffered IO Masters

  .....

- Non-Buffered IO

Parallel and Distributed Computing (CS621)

Question: 19 (Marks: 1) **Attempted Questions: 43** **Total Questions: 50**

The time elapsed between the beginning and the end of its execution on a sequential computer is known as _____.

Choices:

- serial runtime
- parallel runtime
- distributed runtime
- both sequential and parallel runtime

Masters

Parallel and Distributed Computing (CS621)

Question: 20 (Marks: 1) Attempted Questions: 44 Total Questions: 50

Speedup S is computed by _____.

Choices:

- $S = \Theta(n / \log n)$
- $S = \Theta(n \log n)$
- $S = \Theta(\log n)$
- $S = \Theta(\log n)^n$

Masters

Parallel and Distributed Computing (CS621)

Question: 21 (Marks: 1) Attempted Questions: 44 Total Questions: 50

The phenomenon when the speedup become greater than p (number of processing elements) is known as _____ speedup.

Choices:

- linear
- superlinear
- non-linear
- supernon-linear

Masters

Parallel and Distributed Computing (CS621)

Question: 22 (Marks: 1) Attempted Questions: 46 Total Questions: 50

_____ is the product of parallel runtime and the number of processing elements used (p x T_P).

Choices:

Cost

Masters

Overhead

Speedup

Efficiency

Parallel and Distributed Computing (CS621)

Question: 23 (Marks: 1) Attempted Questions: 47 Total Questions: 50

Using fewer than the maximum possible number of processing elements to execute a parallel algorithm is called _____.

Choices:

- scaling up
- scaling down
- scaling out
- scaling in

Masters

Question: 24 (Marks: 1) Attempted Questions: 48 Total Questions: 50

_____ is a special class of distributed data-store which is characterized by the lack of simultaneous updates. Here, the emphasis is more on maintaining a consistent view of things for the individual client process that is currently operating on the data-store.

Choices:

- Sequential model
- Client centric consistency model
- Data consistency model
- Incremental model

Masters

Question: 25 (Marks: 1) Attempted Questions: 49 Total Questions: 50

_____ the data and moving it closer to where it is needed helps to solve the scalability problem.

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Choices:

- Replicating
- Aborting
- Terminating
- Paradigm

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